

## Yes/No Questions Across Chinese Varieties and English

### Abstract

This article provides a morphosyntactic and semantic analysis of nine yes/no question constructions across English and Chinese varieties (e.g., Mandarin, Yixing, Nanjing, Suzhou, Hefei, Kunming, Liangshan, many of which are understudied), and shows that the apparently complex crosslinguistic variation is in fact complete systematic variation generated by just three binary parameters. The first parameter is whether or not the question involves a polar Disjunction Phrase (DisjP) in syntax. The questions with a polar DisjP vary in two more parameters: whether there is overt or covert *wh*-movement, and whether the DisjP is realised as a single Vocabulary Item or multiple. Our study has two theoretical implications. The first is for question semantics and typology: we provide syntactic and morphological evidence for two types of interrogative C<sup>0</sup>s in Hamblin's (1973) question semantics; we also argue that the yes-no questions without DisjP do not involve any *wh*-element, and thus no (c)overt *wh*-movement or even genuine *wh*-in-situ, thus enriching the typology of questions. Our second theoretical contribution lies in theories of parametric variation. The three parameters proposed in this article have three different sources, all of which should be taken into account in parametric theories: linearization, phonological realization of features during Lexical Insertion, and the (un)availability of features in the lexicon.

**Keywords:** yes/no questions; polar Disjunction Phrase; *wh*-movement; parametric variation; Chinese; English

## 1. Introduction

Yes/no questions (i.e., questions that can be answered by ‘yes’ or ‘no’) uniquely sit at the intersection of two important topics in linguistic theory—alternatives and questions. There is variation in both the presentation of alternatives and question formation, leading us to expect wide variation in yes/no question formation. Within a language, we may see yes/no questions differ in whether they involve polarity disjunction: for example, in English, some yes/no questions involve a polarity disjunction phrase (*or not*), while others don’t. Across languages, there is well-known binary variation in *wh*-question formation: the *wh*-phrase moves overtly in some languages but covertly in others. Since these two properties (i.e. alternatives and questions) intersect in yes/no questions, we may expect yes/no questions to have rich variation within and across languages. This paper examines this rich variation in yes/no questions in English and the Chinese language family (including Mandarin and other dialects), and shows that they have the complete parametric variation we expect from the paradigm. Furthermore, they vary in a third dimension: if the yes/no question involves a polarity disjunction, whether that disjunction is realised as a single Vocabulary Item (VI) or multiple. Thus, our morphosyntactic and semantic theories of yes/no questions allow us to find order in the apparently noisy data: the apparently diverse yes/no question types across English and the Chinese dialects are in fact complete systematic variation generated by just three binary parameters. This is significant for the understanding of question semantics and typology, where we argue that there are at least three types of questions and two types of interrogative  $C^0$ s, in support of Hamblin (1973). Our work also bears on parametric variation in Minimalism

by arguing that a combination of two major approaches, the Borer-Chomsky Conjecture (Baker 2008; Borer 2013; Roberts 2019) and the externalisation approach (Berwick and Chomsky 2016; Chomsky 2019; Chomsky et al. 2023) together accounts for the variation of yes/no questions we observe; additionally, they interact with each other to generate the apparent complexity of variation.

We first introduce the theoretical background on each binary variation using basic data, beginning with polarity disjunction. In English, yes/no questions may and may not involve *or not*:

- (1)a. Does John like coffee or not?
- b. Does John like coffee?
- (2)a. I wonder whether John likes coffee or not.
- b. I wonder whether John likes coffee.

Contrary to the claim that these two types of yes/no questions are equivalent, and there is silent *or not* when we don't hear it (e.g. Larson 1985; Romero and Han 2004), it has been observed that these two types of yes/no questions sometimes have different distributions (e.g., Karttunen 1977a; Bolinger 1978; Huddleston 1994; Eckardt 2007; Rawlins 2008; Biezma 2009; Biezma and Rawlins 2012). For example, Karttunen observed that only yes/no questions without *or not* can be the complement of *doubt*:

- (3)a. #I doubt whether John likes coffee or not. *ALTQVN*
- b. I doubt whether John likes coffee. *POLQ*

Evidence like this has led to the proposal that there are two different types of yes/no questions: alternative questions with a disjunction phrase (DisjP) *or not* that contrasts in

polarities ((1a) and (2a), which we call *ALTQVN* following Biezma and Rawlins 2012); and questions without any DisjP ((1b) and (2b), which we call *POLQ*) (e.g., Hamblin 1973; Karttunen 1977b; Rawlins 2008; Biezma 2009; Pruitt and Roelofsen 2011; Biezma and Rawlins 2012; Farkas and Roelofsen 2012; Uegaki 2023).

There are two different types of approaches to the contrast between (3a&b). The first was that the *ALTQVN* in (3a) denotes a doubleton set {John likes coffee, John doesn't like coffee}, whereas the *POLQ* in (3b) denotes a singleton set {John likes coffee}, just like a declarative *that*-clause, and *doubt* selects for singleton sets (Biezma and Rawlins 2012). The second approach adopted the Hamblin semantics of questions that both the *ALTQVN* and the *POLQ* denote a doubleton set, but the *POLQ* highlights only one proposition {John likes coffee}, whereas the *ALTQVN* highlights two {John likes coffee, John doesn't like coffee} (Pruitt and Roelofsen 2011; Farkas and Roelofsen 2012; Uegaki 2023). *Doubt* requires its complement to highlight exactly one proposition.

The Hamblin question semantics followed by the second approach has an important consequence: there are two different interrogative  $C^0$ s, occurring in *ALTQVNs* and *POLQs* respectively. This is based on Hamblin disjunction semantics, where the TP in *ALTQVNs* denotes a doubleton set (i.e., {John likes coffee, John doesn't like coffee} in (3a); Alonso-Ovalle 2005; Simons 2005; Alonso-Ovalle 2006), whereas the TP in *POLQs* denotes a singleton set (i.e., {John likes coffee} in (3b)). To derive the doubleton meaning of these questions, we thus need two different interrogative  $C^0$ s: the  $C^0$  in *ALTQVNs* is an identity function (same as the  $C^0$  in *wh*-constituent questions), whereas the  $C^0$  in *POLQs* turns a singleton set into a doubleton. This proposal was based on

English, where the two interrogative C<sup>0</sup>s look the same. This drew criticism about this approach—that is, across languages, we do not see morphological evidence of two different C<sup>0</sup>s (Biezma and Rawlins 2015).

This paper supports the Hamblin semantic analysis with a close cross-dialectal examination of two Chinese varieties, where the two C<sup>0</sup>s are realised differently. We will not only use the ‘doubt’ test, but also develop new diagnostics to show that two apparently similar yes/no question constructions in the Chinese varieties are in fact ALTQVNs and POLQs respectively. This is the first binary parameter we investigate.

Specifically, we will contrast *fe/me*-marked questions (hereafter *fe/me*-questions) in Chinese Yixing dialect (4a) with *ma*-marked questions (hereafter *ma*-questions) in Mandarin (4b).

- |       |                                       |        |       |        |     |                               |
|-------|---------------------------------------|--------|-------|--------|-----|-------------------------------|
| (4)a. | Zhangsan                              | huexi  | qie   | kafei  | fe? | Yixing <i>fe/me</i> -question |
|       | Zhangsan                              | like   | drink | coffee | FE  |                               |
|       | ‘Does Zhangsan like to drink coffee?’ |        |       |        |     |                               |
| b.    | Zhangsan                              | xihuan | he    | kafei  | ma? | Mandarin <i>ma</i> -question  |
|       | Zhangsan                              | like   | drink | coffee | MA  |                               |
|       | ‘Does Zhangsan like to drink coffee?’ |        |       |        |     |                               |

These two types of yes/no questions appear to be very similar because they are both marked by a sentence-final particle—*fe* (or *me*, depending on the aspect of the clause), and *ma*. But we will show that Mandarin *ma*-questions are POLQs and do not involve a DisjP in the syntax, whereas Yixing *fe/me*-questions are ALTQVNs and involve a DisjP. The C<sup>0</sup> in ALTQVNs is null in Yixing and Mandarin, whereas the C<sup>0</sup> in POLQs is *ma* in Mandarin.

ALTQVNS can be analysed on a par with general alternative questions such as (5). The only difference between them is that the DisjP in ALTQVNS disjoins the positive and negative polarities, whereas the DisjP in alternative questions may disjoin other constituents.

(5) I wonder whether John likes coffee or tea.

If we follow Larson (1985), Romero and Han (2004), Wu (2022), and others' view that alternative questions and ALTQVNS involve a *wh*-element (e.g., *whether* in English) that moves from the left edge of the DisjP to Spec, CP, then we can unify alternative questions with *wh*-constituent questions. This leads us to ask whether the yes/no questions in Chinese are *wh*-questions. We will show that the ALTQVNS (i.e., Yixing *fe/me*-questions) have *wh*-properties, but POLQS (i.e., Mandarin *ma*-questions) don't. This has the consequence that the  $C^0$  in ALTQVNS triggers *wh*-movement, whereas the  $C^0$  in POLQS doesn't.

The lack of *wh*-movement in *ma*-questions might remind us of the lack of *wh*-movement in genuine *wh*-in-situ constructions (e.g., English superiority-violating questions), where a *wh*-in-situ question has *wh*-question meaning via alternative computation, unselective binding or choice functions, but we will show that *ma*-questions do not involve genuine *wh*-in-situ (i.e., they are not subject to intervention effects). Our work on *ma*-questions thus enriches the typology of questions, suggesting that there are at least three types of questions: (a) questions with (overt or covert) *wh*-movement (e.g., standard *wh*-constituent-questions, Yixing *fe/me*-questions and Mandarin *bu/mei*-questions); (b) genuine *wh*-in-situ questions without any *wh*-movement (e.g., English

superiority-violating questions and Japanese *wh*-in-situ; Kotek 2018); and (c) yes/no questions without *wh*-movement or *wh*-in-situ (e.g., Mandarin *ma*-questions).

We will not only argue that Yixing *fe/me*-questions involve *wh*-movement, but also that this movement is overt. This may be surprising because *wh*-constituent movement in the Chinese languages has been traditionally assumed to be covert by default since C.-T. J. Huang (1982). But we will show that not all yes/no questions in Chinese involve overt *wh*-movement; some can involve covert *wh*-movement as well, such as A-not-A questions in Mandarin and *a/ke*-questions in other Chinese varieties. This is the second parameter of variation—if a yes/no question involves *wh*-movement, whether this movement is overt or covert. A-not-A questions typically reduplicate the first syllable of the main predicate and insert negation after the reduplicant:

(6) Zhangsan xi bu xihuan he kafei?

Zhangsan RED NEG like drink coffee

‘Does Zhangsan like to drink coffee?’

We will argue that A-not-A questions involve a DisjP, just like *fe/me*-questions, but their DisjP (i.e., RED NEG) appears sentence-medially, rather than at the clause edge as in *fe/me*-questions. We still observe effects of cross-clausal movement in A-not-A questions, suggesting that they involve covert *wh*-movement, in support of C.-T. J. Huang (1991).

Having introduced two parameters underlying the cross-linguistic variation of yes/no questions—whether they involve a DisjP, and whether they involve overt or covert *wh*-movement—the third parameter is if they involve a DisjP, whether that DisjP is

phonologically realised as a single Vocabulary Item in the sense of Distributed Morphology (Halle and Marantz 1993; Marantz 2006; Embick 2010; 2015 a.o.) or as multiple VIs. We will argue that Mandarin A-not-A and Yixing *fe/me* realise the DisjP as multiple VIs. They will be contrasted with Nanjing, Suzhou (including Taicang and Changshu), Hefei, and Kunming dialects, whose yes/no questions insert a single particle such as *a* (in Nanjing and Suzhou) and *ke* (in Hefei and Kunming) before the VP (hereafter *a/ke*-questions):

(7) Zhangsan a xihuan huo kafei? *Nanjing*

Zhangsan A xihuan drink coffee

‘Does Zhangsan like to drink coffee?’

We will argue that *a* and *ke* are the DisjP in these questions, and a single VI. We will further show that the interaction between the second and third parameters predicts the existence of another yes/no question construction, where the DisjP is realised as a single VI like *a/ke*, and it undergoes overt *wh*-movement as in Yixing *fe/me*-questions. These are precisely Liangshan *bo*-questions, whose question marker *bo* is a single VI realising DisjP and undergoes overt *wh*-movement to the sentence-final position.

(8) Zhangsān xihuān hē kāfēi bō? *Liangshan*

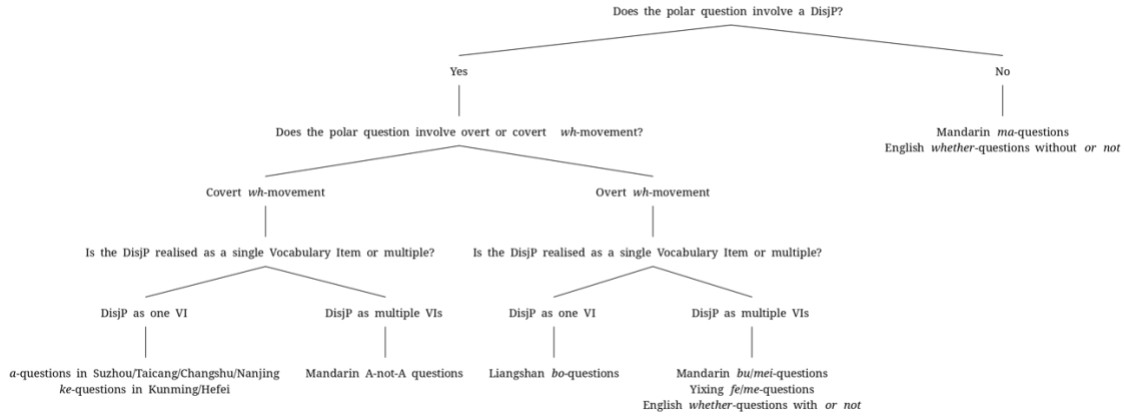
Zhangsan xihuan drink coffee BO

‘Does Zhangsan like to drink coffee?’



Below is a summary of the three binary parameters of yes/no questions that we will examine in detail in the remainder of this paper.<sup>1</sup>

### (9) Parameters of Yes/no Questions



We demonstrate the rich diversity of yes/no questions in the Chinese varieties and English using Roberts' (2019) parameter hierarchy. With only three binary parameters at play, we observe complex but systematic cross-linguistic variation of yes/no questions. Seemingly noisy data can be reduced to a finite number of parameters, consonant with parametric variation in the Minimalist Program.

<sup>1</sup> Some might assume implicitly that there can only be parametric variation across different languages. But some parameters in our study describe variation within a language: for example, Mandarin and English have both ALTQVNs and POLQs, thus our first parameter is an intra-linguistic one. But some versions of the Borer-Chomsky Conjecture (e.g., Chomsky 2001) would allow parametric variation within a language, since they claimed that variation is due to different features carried by a lexical item. Previous studies have also implied that there can be parametric variation within a certain language, such as the co-existence of overt *wh*-movement, covert *wh*-movement and genuine *wh*-in situ in English (Pesetsky 2000; Kotek 2016).

The organisation of this paper follows the parametric hierarchy in (9) closely. We begin with a discussion of the first parameter (i.e., whether a yes/no question involves a DisjP or not, making it a POLQ or an ALTQVN) in section 2, where we present our core assumptions about the syntax of yes/no questions and argue that Yixing *fe/me*- and Mandarin *bu/mei*-questions are ALTQVNs, whereas Mandarin *ma*-questions are POLQs. We will also discuss the implications this has for Hamblin semantics of questions and the typology of questions. Section 3 moves on to the second parameter (i.e., whether a yes/no question involves overt or covert *wh*-movement), and proposes a new analysis of A-not-A questions that differs from but draws insights from the main approaches in the literature, Zhu (1985); C.-T. J. Huang (1991); C.-T. J. Huang et al. (2009). Section 4 examines the third parameter (i.e., whether the DisjP is realised as a single VI or multiple) based on Nanjing, Suzhou, Hefei, and Kunming. This section also shows that an interaction of the parameters makes an important prediction that is borne out in another Chinese dialect, Liangshan. Section 5 concludes with a discussion of the two main theoretical contributions of the paper: one for question semantics and typology, and the other for parametric theories. Specifically, our parametric hierarchy supports three types of theories of parametric variation in the Minimalist Program, with each parameter supporting one type of theory.

## **2. Whether the yes/no question involves a DisjP**

This section focuses on three types of yes/no questions in the Chinese family, including two in Mandarin and one in Yixing Chinese, a variety of Chinese Wu spoken in Jiangsu Province, China. While all three types insert a sentence-final particle (SFP) at the end of

a declarative, we will show that they represent two different ways of deriving a yes/no question, with and without a DisjP. We begin this section by first comparing two of these three types of questions, Mandarin *ma*-questions and Yixing *fe/me*-questions in section 2.1, and argue that the former don't involve a DisjP, but the latter do. Then we provide their syntactic analyses in section 2.2, and argue that Mandarin *ma*-questions involve a movement from C<sup>0</sup> to SpeechAct<sup>0</sup> in section 2.3. In section 2.4 we add the third type of question, Mandarin *bu/mei*-questions, to the discussion, which we argue are a hybrid of *ma*- and *fe/me*-questions.

## 2.1. Yixing *fe/me*-questions involve a polar DisjP, but Mandarin *ma*-questions don't

In Mandarin Chinese, a yes/no question can be created by adding the SFP *ma* to the end of a declarative sentence ((4a) is repeated below):

(10) Zhangsan      xihuan   he   kafei.

Zhangsan      like      drink coffee

‘Zhangsan likes to drink coffee.’

(4)a. Zhangsan   xihuan   he   kafei   ma?

Zhangsan   like      drink coffee   MA

‘Does Zhangsan like to drink coffee?’

In Yixing, a yes/no question can be created by adding *fe* or *me* to the end of a declarative sentence. *Me* is used in sentences with a perfective viewpoint aspect (i.e. sentences that ‘present a situation as a single whole’; Smith 1997:103), and *fe* is used elsewhere. An

additional SFP *a* can follow them, and can fuse with *fe* or *me* to turn into *fa* and *mang* respectively.

(11) Zangsa    huexi qie    kafi    de.

Zhangsan   like   drink coffee   DE

‘Zhangsan likes to drink coffee.’

(12) Zangsa    huexi qie    kafi    {fe/fe a/fa}?

Zhangsan   like   drink coffee   FE/FE A/FA

‘Does Zhangsan like to drink coffee?’

(13) Zangsa    jinzaio    qie    kafi    ge.

Zhangsan   today   drink   coffee   GE

‘Zhangsan drank coffee today.’

(14) Zangsa    jinzaio    qie    kafi    {me/me a/mang}?

Zhangsan   today   drink coffee   ME/ME A/MANG

‘Did Zhangsan drink coffee today?’

We think that the sentence-final *fe* and *me* in yes/no questions are genuine negations for two reasons. First, they are identical with the negation words in the language. These negation words are also sensitive to the aspect of the sentence: *me* is the negation used with the perfective aspect, and *fe* is the negation elsewhere:

(15) a. Zangsa    fe huexi qie    kafi.

Zhangsan   FE like   drink coffee

‘Zhangsan doesn’t like to drink coffee.’

b. Zangsa    jinzao    me    qie    kafi.

Zhangsan    today    MEDrink coffee

‘Zhangsan didn’t drink coffee today.’

Second, like negation, the SFPs *fe* and *me* are incompatible with focus and commitment particles like *de* and *lie*. We do not provide an analysis of *de* and *lie*, but just note here that *fe* and *me* have the same distribution as negation with respect to *de* and *lie*. *De*<sup>2</sup> or *lie* is required in positive declaratives but disallowed in negative ones:

(16) a. Zangsa    huexi qie    kafi    de.

Zhangsan    like    drink    coffee    DE

‘Zhangsan likes to drink coffee.’

b. Zangsa    qie-le    sa    bai    kafi    lie

Zhangsan    drink-PERF    three cup    coffee    LIE.

‘Zhangsan has drunk three cups of coffee.’

(17) a. Zangsa    fe    huexi    qie    kafi    (\*de).

Zhangsan    NEG    like    drink    coffee    DE

‘Zhangsan doesn’t like to drink coffee.’

b. Zangsa    me    qie-le    sa    bai    kafi    (\*lie).

Zhangsan    NEG    drink-COMPL    three cup    coffee    LIE

‘Zhangsan didn’t drink three cups of coffee.’

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<sup>2</sup> *De* in Yixing is not equivalent to the particle *de* in Mandarin. The Yixing particle corresponding to the Mandarin *de* is *ge*, as in (13).

*De* and *lie* are disallowed in *fe/me*-questions; instead, only *a* can co-occur with *de* and *lie*.

We take the same distribution of *fe/me* and negation to indicate that *fe* and *me* are negation.

- (18) a. Zangsa      zenglao      huexi      qie      kafi      de { \*fe/\*fe a/\*fa/a } ?  
           Zhangsan    really      like      drink      coffee    DE    FE/FE A/FA ?  
           ‘Is it true that Zhangsan really likes to drink coffee?’
- b. Zangsa      yijing      qie-le      sa      bai      kafei      lie { \*me/\*me a/\*mang/a } ?  
           Zhangsan    already    drink-PERF    three cup    coffee    LIE    ME/ME A/MANG/A  
           ‘Did Zhangsan already drink three cups of coffee?’

While Mandarin *ma* is not currently used as negation in the language, there is evidence that it evolved from negation historically (e.g., Aldridge 2011). This makes Mandarin *ma*-questions and Yixing *fe/me*-questions look very similar: they are all created by inserting (what is currently or historically) a negation particle at the end of a declarative. But we will argue using four diagnostics that they are very different: Yixing *fe/me*-questions are ALTQVNS because they involve a polar DisjP equivalent to English *or not*, whereas Mandarin *ma*-questions are POLQs because they don’t involve any DisjP or *wh*-movement. As we present those four tests in Mandarin and Yixing, we will also use them on English *whether*-questions to show that *whether*-questions with overt *or not* are ALTQVNS, whereas those without overt *or not* are POLQs.

First, *ma*-questions can be answered with *shi de*, a cleft construction that we translated as ‘It is so’ (C.-T. J. Huang et al. 2009), whereas *fe/me*-questions can’t. Instead, the verb is required in the polar answer.

(19) A: Zhangsan xihuan he kafei ma?

Zhangsan like drink coffee MA

‘Does Zhangsan like to drink coffee?’

B: shi de.

COP DE

‘It is so.’

(20) A: Zhangsan huexi qie kafi {fe/fe a/fa}?

Zhangsan like drink coffee FE/FE A/FA

‘Does Zhangsan like to drink coffee?’

B1: #shi ge.

COP GE

‘It is so.’

B2: huexi de.

like DE

‘Yes.’

Following Roelofsen and van Gool (2010), Pruitt and Roelofsen (2011), Roelofsen and Farkas (2015), and Mohammadi and Romero’s (2024) analysis of ‘yes’ and ‘no’ answers in English and Farsi, we think that the ‘so’-answer requires the question to highlight only one proposition. According to these proposals, the highlighted value of a question only includes the answers that are mentioned overtly in the syntactic form of the TP. Thus, POLQs highlight only one proposition (i.e., the TP), whereas ALTQVNs highlight two propositions at the same time (i.e., the positive proposition and the negative). We thus

take the contrast between (19) and (20) to indicate that *fe/me*-questions involve a DisjP, whereas *ma*-questions don't.

In parallel, English questions with *or not* cannot be answered using *so*, whereas those without *or not* can, suggesting that the former are ALTQVNs and the latter are POLQs:

(21) A: Does John like to drink coffee or not? *ALTQVN*

B: #I guess so / #It seems so / #Maybe so.

(22) A: Does John like to drink coffee? *POLQ*

B: I guess so / It seems so / Maybe so.

The second diagnostic is based on Karttunen's (1977) observation in English that *doubt* can only take *that*-declaratives and yes/no questions without *or not*, but not yes/no questions with *or not*:

(23) a. John doubts that Marie stole the money.

b. John doubts whether Marie stole the money. *POLQ*

c. #John doubts whether Marie stole the money or not. *ALTQVN*

As discussed in section 1, we follow Pruitt and Roelofsen (2011), Farkas and Roelofsen (2012) and Uegaki (2023) in assuming that *doubt* requires its complement to highlight exactly one proposition. POLQs highlight only one proposition, whereas ALTQVNs highlight two, the positive and the negative propositions.



In parallel, ‘doubt’ in Mandarin and Yixing can only take a single proposition (24), not a disjunction of two propositions (25).<sup>3</sup> There are two ‘or’s in Mandarin, *haishi*, the ‘or’ associated with ‘whether’, and *huozhe*, the ‘or’ associated with ‘either’. *Haishi* is not possible in the clause embedded by ‘doubt’ (25), but *huozhe* is (26). Crucially, (26) only has the reading that Zhangsan suspects that one of Lisi and Wangwu stole the money.

(24) Zhangsan huaiyi Lisi tou-le qian.

Zhangsan suspect Lisi steal-PERF money

‘Zhangsan suspects Lisi stole the money.’

(25) #Zhangsan huaiyi Lisi haishi Wangwu tou-le qian.

Zhangsan suspect Lisi or<sub>whether</sub> Wangwu steal-PERF money

Intended: ‘Zhangsan suspects whether Lisi or Wangwu stole the money.’

(26) Zhangsan huaiyi Lisi huozhe Wangwu tou-le qian.

Zhangsan suspect Lisi or<sub>either</sub> Wangwu steal-PERF money

‘Zhangsan suspects that either Lisi or Wangwu stole the money.’

‘Doubt’ cannot embed *fe/me*-questions (27), even though *fe/me*-questions can generally be embedded (e.g. under ‘know’, (28)). This suggests that *fe/me*-questions are ALTQVNS. It is worth mentioning that while *fe/me*-questions can be generally embedded, those with the additional *a* can’t be, a fact we will discuss in depth later in section 2.3.

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<sup>3</sup> *Huaiyi* actually has the meaning of English *suspect*, not *doubt* because the subject of *huaiyi* is inclined to believe that the complement proposition of *huaiyi* is true, not false. But since *huaiyi* requires highlighting a single proposition, we take it to behave like *doubt* in this aspect.

(27) #Zangsa huani Lisi huexi qie kafi {fe/fe a/fa}.<sup>4</sup>

Zhangsan suspect Lisi like drink coffee FE/FE A/FA

Intended: ‘Zhangsan suspects whether Lisi likes to drink coffee.’

(28) wo xiaoze lao Zangsa huexi qie kafi {fe/\*fe a/\*fa}.

I know LAO Zhangsan like drink coffee FE/FE A/FA

‘I know whether Zhangsan likes to drink coffee.’

‘Doubt’ cannot embed *ma*-questions, either, which may at first appearance suggest that *ma*-questions involve a DisjP:

(29) \*Zhangsan huayi Lisi xihuan he kafei ma.

Zhangsan suspect Lisi like drink coffee MA

Intended: ‘Zhangsan suspects whether Lisi likes to drink coffee.’

But we think this is because *ma*-questions cannot be embedded at all, even under verbs that can otherwise embed questions, such as ‘know’:

(30) \*wo zhidao Lisi xihuan he kafei ma.

I know Lisi like drink coffee MA

Intended: ‘I know whether Lisi likes to drink coffee.’

Thus, *fe/me*-questions cannot be embedded under ‘doubt’ because they are ALTQVNs and involve a DisjP, whereas Mandarin *ma*-questions and Yixing *a*-questions cannot be

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<sup>4</sup> This sentence can be interpreted as a yes/no question, with both these meanings ‘Does Zhangsan suspect that Lisi likes to drink coffee, or does he suspect that Lisi doesn’t like to drink coffee?’ and ‘Does Zhangsan suspect that Lisi likes to drink coffee, or does he not suspect that Lisi likes to drink coffee?’ We will discuss these interpretations in section 2.2.

embedded in general. Section 2.3 will provide an account of the embeddability of *fe* and *me* but the unembeddability of *ma* and *a*.

Our third diagnostic is based on Rawlins' (2008) observation that English ALTQVNs can be used as unconditionals, whereas POLQs cannot:

- (31) a. #Whether John likes to drink coffee, we will meet at Starbucks. *POLQ*  
 b. Whether John likes to drink coffee or not, we will meet at Starbucks. *ALTQVN*

In parallel, *fe/me*-questions can occur in unconditionals, suggesting they are ALTQVNs.

Sentence-final *a* cannot occur here, which again we will explain in section 2.3.

- (32) fegue      ni      mingzao      lai      wo      guo      qie      va      {fe/\*fe a/\*fa},  
 no.matter   you   tomorrow   come my   home have meal   FE/FE A/FA  
 wo   jinzao   zongguai   yao   ma   maduo   cai      de.  
 I      today   anyway   will   buy   many   groceries   DE  
 'Whether you're coming for a meal or not, I'll buy many groceries today anyway.'

*Ma*-questions cannot be embedded in unconditionals, but this does not indicate they are POLQs because as we saw in (30), *ma*-questions are generally unembeddable.

- (33) \*buguan      ni      mingtian      lai      wo      jia      chi      fan      ma,  
 no.matter   you   tomorrow   come my   home have meal   MA  
 wo   jintian   dou   yao   mai   henduo   cai      de.  
 I      today   DOU   will   buy   many   groceries   DE

Intended: 'Whether you're coming for a meal or not, I'll buy many groceries.'

Our final diagnostic of a polar DisjP involves negative yes/no questions. *Ma*-questions can involve a negative proposition (C.-T. J. Huang et al. 2009), but *fe/me*-questions cannot. Instead, only *a* appears in negative yes/no questions in Yixing:

(34) a. Zhangsan bu he kafei ma?

Zhangsan NEG drink coffee MA

‘Does Zhangsan not drink coffee?’

b. Zhangsan mei he kafei ma?

Zhangsan NEG drink coffee MA

‘Did Zhangsan not drink coffee?’

(35) a. Zangsa fe qie kafi { \*fe/\*fe a/\*fa/a }?

Zhangsan NEG drink coffee FE/FE A/FA/A

Intended: ‘Does Zhangsan not drink coffee?’

b. Zangsa me qie kafi { \*me/\*me a/\*mang/a }?

Zhangsan NEG drink coffee ME/ME A/MANG/A

Intended: ‘Did Zhangsan not drink coffee?’

These negative yes/no questions check the truth and falsity of a negative proposition and can license negative polarity items (NPIs), and therefore have what Romero and Han (2004) called the inner negation reading, equivalent to their English translations. Like their English translations, they can involve bias, but do not have to. We observe that in parallel, English POLQs do not allow negative propositions, either, whereas ALTQVNs do:

(36) a. \*Does John not drink coffee or not?

b. Does John not drink coffee?

We therefore think that the yes/no question's compatibility with a negative proposition can be a diagnostic of whether it involves a polar DisjP: Yixing *fe/me*-questions do, but Mandarin *ma*-questions don't.

To sum up, we have presented four tests, answerability by 'so', embeddability by 'doubt', embeddability in unconditionals, and negative yes/no questions, that all suggest that Yixing *fe/me*-questions are ALTQVNS and involve a polar DisjP, whereas Mandarin *ma*-questions are POLQS and don't involve any polar DisjP.

It is worth mentioning that Mandarin and Yixing have yes/no questions with a transparent DisjP disjoined by Mandarin *haishi* or Yixing *asi* 'or<sub>whether</sub>'. 'Or<sub>whether</sub>' can disjoin phrases of various sizes, such as DPs, VPs and clauses, and 'or<sub>whether</sub>' can be silent. When predicates of opposite polarities are disjoined, we get a yes/no question meaning:

(37) Zhangsan he kafei (haishi) bu he?

Zhangsan drink coffee or<sub>whether</sub> NEG drink

'Does Zhangsan drink coffee or not?'

These questions pass all our tests of polar DisjP, and since they clearly involve a DisjP, we will not discuss them further. Section 2.2.3 will argue that *fe/me*-questions are not derived from these transparently alternative questions plus ellipsis.

## 2.2. Syntax of *fe/me*- and *ma*-questions

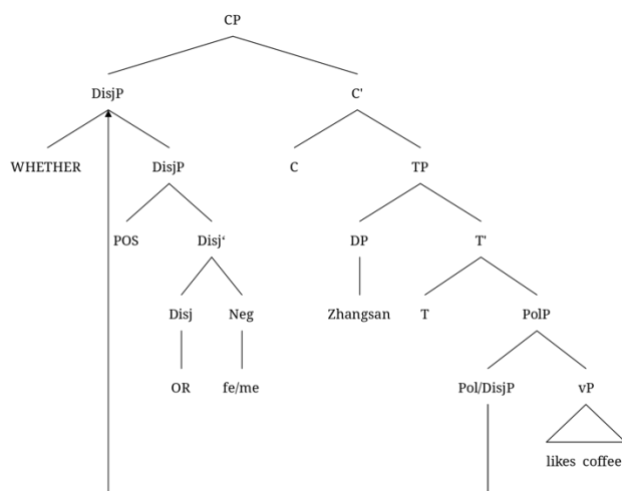
Having shown that *fe/me*-questions are ALTQVNs but *ma*-questions are POLQs, this subsection provides our syntactic analyses of them, and then excludes an ellipsis-based analysis of *fe/me*-questions.

### 2.2.1. *Fe/me*-questions

Since *fe* and *me* in Yixing are genuine negation, we propose they are the negation in the polar DisjP parallel to *not* in English *or not*. In this polar DisjP, only the negative polarity is pronounced, but the positive polarity and ‘or’ are not (i.e., phonologically covert, leading to  $\langle POS \rangle \langle OR \rangle fe/me$ , where silent VIs are in  $\langle \rangle$ ). This is not implausible because as discussed at the end of subsection 2.1, in general, ‘or’ in the Chinese languages can be silent.

As the polar DisjP,  $\langle POS \rangle \langle OR \rangle fe/me$  is base-generated in  $Pol(arity)^0$ , hence its sensitivity to aspect, a general property of negation in Chinese languages. This also accounts for *fe* and *me*’s incompatibility with negative propositions: both negation in the negative proposition and the polar DisjP need to occupy  $Pol^0$ . When  $Pol^0$  is already occupied by negation, there is no place for the polar DisjP. While *fe* and *me* are base-generated sentence-medially in  $Pol^0$ , we argue that they appear sentence-finally in yes/no questions because as a part of the polar DisjP, they have been pied-piped by the *wh*-movement of a covert ‘whether’:

(38)



The DisjP appears sentence-finally because complementisers in Chinese generally follow their syntactic sister. There are many proposals for why this is, and they can all explain why the DisjP appears sentence-finally here. We won't go into detail, but just mention two possibilities. One is Pan's (2022) suggestion of massive fronting of the entire TP, and the other is Chomsky's (2019) suggestion that specifiers (i.e., Spec, CP here) can be merged on either side of the head because Merge in narrow syntax does not specify linear order.

We provide two pieces of evidence that *fe/me*-questions involve *wh*-movement. First, sentence-final *fe/me* can have long-distance dependencies across finite clauses and be interpreted as an embedded yes/no question:

(39) Zangsa huani Lisi huexi qie kafi fe?

Zhangsan suspect Lisi like drink coffee FE

✓Embedded reading: 'Does Zhangsan suspect that Lisi likes to drink coffee, or does Zhangsan suspect that Lisi doesn't like to drink coffee?'

✓Matrix reading: ‘Does Zhangsan suspect that Lisi likes to drink coffee, or does

Zhangsan not suspect that Lisi likes to drink coffee?’

The embedded reading is the result of *wh*-movement from the embedded Pol ((40); silent morphemes are in  $\langle \rangle$ ), whereas the matrix reading is the result of *wh*-movement from the matrix Pol (40).

(40) a. Zangsa huani [<sub>CP</sub> Lisi [<sub>PolP</sub> *t*<sub>i</sub> huexi qie kafi]] [<sub><WHETHER></sub> <sub><POS></sub> <sub><OR></sub> *fe*]<sub>i</sub>?

b. Zangsa [<sub>PolP</sub> *t*<sub>i</sub> huani [<sub>CP</sub> Lisi huexi qie kafi]] [<sub><WHETHER></sub> <sub><POS></sub> <sub><OR></sub> *fe*]<sub>i</sub>?

Second, this long-distance dependency cannot cross an island (e.g., a relative clause), suggesting that this *wh*-movement is island-bound:

(41) \**ni* pongdao-le [<sub>Relative Clause</sub> huexi qie kafi ge wosangzi] *fe*?

you meet-PERF like drink coffee C student FE

Intended Embedded Reading: ‘Did you meet the student who likes to drink coffee, or did you meet the student who doesn’t like to drink coffee?’

Sentence (41) is ungrammatical because the covert ‘whether’ associated with the embedded polar DisjP cannot move out of the relative clause. *Fe* cannot be associated with the matrix clause because the matrix predicate is perfective.

(42) \**ni* pongdao-le [<sub>Relative Clause</sub> [<sub>PolP</sub> *t*<sub>i</sub> huexi qie kafi ge wosangzi] [<sub><WHETHER></sub> <sub><POS></sub> <sub><OR></sub> *fe*]<sub>i</sub>?

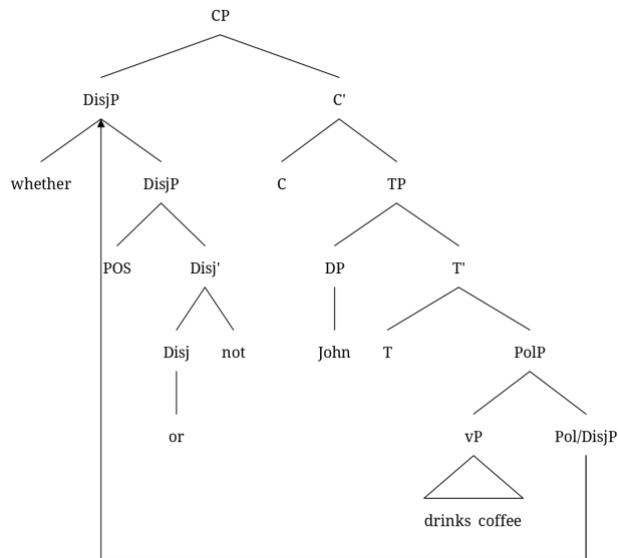
The pied-piping of the polar DisjP by the covert ‘whether’ in Yixing parallels what Wu (2022) argued to be the pied-piping of *or not* by *whether* in English. Building on Larson’s (1985) and Han and Romero’s (2004) idea that *whether* is a *wh*-word and moves, Wu further argued that *whether* pied-pipes the polar DisjP optionally:



- (43) a. I don't know whether John likes coffee or not. *No pied-piping*  
 b. I don't know whether or not John likes coffee. *Pied-piping*

Wu assigned the following structure to (43b), which is identical to the structure we propose for Yixing *fe/me*-questions. The only difference between them is that English *whether*'s pied-piping is optional, whereas Yixing covert 'whether' always pied-pipes the polar DisjP.

(44)



### 2.2.2. *Ma*-questions

In contrast to Yixing *fe/me*-questions, Mandarin *ma*-questions do not involve any DisjP, and nothing moves there. *Ma* is base-generated in C<sup>0</sup> as a complementiser. Unlike *fe/me*-questions, *ma*-questions cannot involve long-distance dependencies across finite clauses. The following sentence can only be interpreted as a matrix yes/no question, and lacks the embedded reading:

(45) Zhangsan huaiyi [CP Lisi xihuan he kafei] ma?

Zhangsan doubt Lisi like drink coffee MA

\*Embedded reading: ‘Does Zhangsan suspect that Lisi likes to drink coffee or does Zhangsan suspect Lisi doesn’t like to drink coffee?’

✓Matrix reading: ‘Does Zhangsan suspect that Lisi likes to drink coffee or does Zhangsan not suspect Lisi likes to drink coffee?’

This suggests that no question particle has moved across the finite clause. Because *ma* cannot be embedded in general, it cannot appear in any island; thus, we cannot test island effects on *ma*-questions.

Before presenting the syntactic analysis of *ma*-questions, we introduce the basics of Hamblin semantics of questions that our analysis supports and follows. In order to capture the difference between ALTQVNs and POLQs, Hamblin (1973) proposed two different question operators, which can be considered to be two different interrogative  $C^0$ s for our purposes. The first interrogative  $C^0$  is an identity function that has to combine with an alternative set with more than one member (46). A question is thus an alternative set that raises an issue using pragmatic mechanisms. Assuming that propositions containing a disjunction or a *wh*-constituent like *what* and *who* denote alternative sets, this interrogative  $C^0$  combines with these propositions to create alternative questions (including ALTQVNs) and *wh*-constituent questions. We thus call this  $C^0$  *ALTQVN*  $C^0$ .

(46) Where  $\llbracket \alpha \rrbracket = \{A, B, C \dots\}$  (undefined if a singleton set):

$$\llbracket C_{ALTQVN} \alpha \rrbracket = \llbracket \lambda w. A(w), \lambda w. B(w), \lambda w. C(w) \dots \rrbracket$$

Following Alonso-Ovalle (2005), Simons (2005) and others, we assume that a disjunction denotes an alternative set with its disjuncts as the alternatives. For example, *coffee or tea* denotes {coffee, tea}, and *John likes coffee or not* denotes {John likes coffee, John doesn't like coffee}. Following Wu (2022), *whether* existentially quantifies over a choice-functional trace applying to the alternative set, leading to the meaning  $[\lambda w. \lambda p: \exists f. p = [\lambda w'. \text{John likes } f\{\text{coffee, tea}\} \text{ in } w']]$  for *Does John like coffee or tea?* and  $[\lambda w. \lambda p: \exists f. p = \lambda w'. f\{\text{John likes coffee in } w', \text{John doesn't like coffee in } w'\}]]$  for *Does John like coffee or not?*.

Besides the ALTQVN  $C^0$ , the other interrogative  $C^0$  (which we call *POLQ*  $C^0$ ) has to combine with a singleton set rather than an alternative set with multiple members (47). It turns the content proposition that it combines with into an alternative set with two members, one being the content proposition itself, and the other being its negation. This interrogative  $C^0$  only occurs in POLQs.

(47) Where  $\llbracket \alpha \rrbracket = \{A\}$  (undefined if not a singleton set):

$$\llbracket C_{\text{POLQ}} \alpha \rrbracket = \{\lambda w. A(w), \lambda w. \neg A(w)\} \quad (\text{Hamblin 1973:50})$$

In support of Hamblin's two types of  $C^0$ s, the null  $C^0$  in *fe/me*-questions is ALTQVN  $C^0$ , whereas *ma*, the complementiser in Mandarin *ma*-questions, is POLQ  $C^0$ . Evidence that *ma* is POLQ  $C^0$  is that *ma* can only appear in yes/no questions, but not alternative questions or *wh*-constituent questions:

(48) a. Zhangsan xihuan he kafei haishi cha (\*ma)?

Zhangsan like drink coffee orwhether tea MA?

'Does Zhangsan like to drink coffee or tea?'

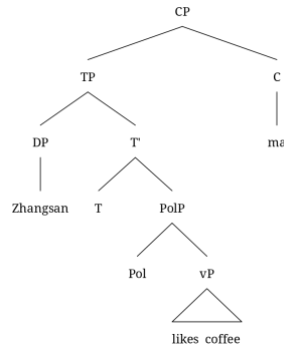
b. Zhangsan xihuan na zhong kafei (\*ma)?

Zhangsan like which kind coffee MA?

‘Which kind of coffee does Zhangsan like?’

This is because *ma* has to combine with a singleton set, and propositions with a DisjP or a *wh*-constituent denote an alternative set with multiple members. The proposition in the singleton set that *ma* combines with can be positive or negative. When the proposition is negative, we get negative yes/no questions. Following is our syntactic analysis of *ma*-questions:

(49)



The lack of *wh*-movement in *ma*-questions might make us think of the lack of *wh*-movement in some genuine *wh*-in-situ constructions (e.g., English superiority-violating questions), but they are different. Genuine *wh*-in-situ constructions are subject to intervention effects by focus-sensitive operators (e.g. *even*, *only*), negation and *every* (50), but *ma*-questions aren't.

(50) a. [Which student]<sub>i</sub> didn't t<sub>i</sub> read which book?

*superiority-obeying*

b. \*[Which book] didn't which student read t<sub>i</sub>?

*superiority-violating*

As we saw in (34a-b), negation can occur in *ma*-questions. So can focus-sensitive operators like *zhi* ‘only’:

(51) Zhangsan zhi xihuan kafei ma?

Zhangsan only like coffee MA

‘Does Zhangsan only like coffee?’

Therefore, *ma*-questions are interesting because they enrich the typology of questions. They suggest that there are at least three types of questions: (a) questions with (overt or covert) *wh*-movement (e.g., standard *wh*-constituent-questions, Yixing *fe/me*-questions, Mandarin *bu/mei*-questions); (b) genuine *wh*-in-situ questions without any *wh*-movement (e.g., English superiority-violating questions, Japanese *wh*-in-situ; Kotek 2018); and (c) yes/no questions without any *wh*-element, and therefore no *wh*-movement or *wh*-in-situ (e.g., Mandarin *ma*-questions). Mandarin *ma* is base-generated in  $C^0$ , but its derivation does not end there. In subsection 2.3 we will argue that it moves to  $\text{SpeechAct}^0$ , a position occupied by *a* in Yixing. This will account for their unembeddability.

### 2.2.3. Against an ellipsis-based analysis of *fe/me*-questions

Our analysis of *fe/me*-questions assumes that *fe* and *me* are part of the polar DisjP. A plausible alternative hypothesis is that *fe/me*-questions involve underlying disjunction of positive and negative propositions by a silent ‘or<sub>whether</sub>’, followed by ellipsis in the second disjunct. The sentence-final *fe/me* is then the negation in the second disjunct:

(52) Zangsa [huexi qie kafi] [fe ~~huexi qie~~ kafi]?

Zhangsan like drink coffee NEG like drink coffee

‘Does Zhangsan like to drink coffee or not?’

This subsection provides three pieces of evidence against this ellipsis-based analysis. The first concerns negating the verb *you* ‘possess’, which in Yixing requires the negation *me*, not *fe*, even if the predicate is imperfective:

- (53) a. Zangsa     you     xu.  
                  Zhangsan   possess   book  
                  ‘Zhangsan has books.’
- b. Zangsa     {\*fe/me}     xu.  
                  Zhangsan   have.NEG   book  
                  ‘Zhangsan does not have books.’

In an alternative question with transparent disjunction of two propositions, *me* is used in the negative proposition, not *fe*:

- (54) Zangsa     [you xu]     [{\*fe/me}     xu]?  
                  Zhangsan   have book     have.NEG   book  
                  ‘Does Zhangsan have books or not?’

If we apply ellipsis to the second disjunct of (54), we would expect *me* to occur sentence-finally. Contrary to this prediction, *fe* but not *me* surfaces. According to our analysis, this is because the negation in the polar DisjP is sensitive to the imperfective aspect.

- (55) Zangsa     you   xu     {fe/\*me}?  
                  Zhangsan   have   book     FE/ME  
                  ‘Does Zhangsan have books or not?’

The second piece of evidence comes from island constraints. Alternative questions involving transparent disjunction in Yixing are not subject to island constraints:<sup>5</sup>

(56) [<sub>subject</sub> zangsa [be livo] asi[fe be livo]] wai niang ni kaixin?

Zhangsan give gift or NEG give gift will make you happy

‘Which will make you happy: Zhangsan giving you a gift or Zhangsan not giving you a gift?’

If *fe/me*-questions are indeed alternative questions plus ellipsis, they should not be sensitive to islands, contrary to fact (41). Also contrast (56) with (57).

(57) \* [<sub>subject</sub> Zangsa be livo fe] wai niang ni kaixin?

Zhangsan give gift FE will make you happy

If (57) is derived by deleting the VP in the second disjunct of (56), then (57) should be fine, contrary to fact. According to our analysis, *fe* must be pied-piped to matrix Spec, CP, and thus cannot occur in embedded clauses, as in (57). *Fe* can occur sentence-finally, as in (58), but it only has the matrix reading.

(58) [<sub>subject</sub> Zangsa be livo] wai niang ni kaixin fe?

Zhangsan give gift will make you happy FE

\*Embedded reading: ‘Which will make you happy: Zhangsan giving you a gift or Zhangsan not giving you a gift?’

---

<sup>5</sup> In parallel, alternative questions with transparent disjunction in Mandarin Chinese are also insensitive to island constraints (C.-T. J. Huang et al. 2009; Holmberg 2016).

✓Matrix reading: ‘Will Zhangsan’s giving you a gift make you happy or not happy?’

According to our analysis, there is a covert *wh*-operator in *fe/me*-questions. It can only move from the matrix Pol, but not from the Pol embedded in the subject because it is subject to the subject island.

Finally, the third piece of evidence comes from the adjectival marker *lao* in Yixing. In declarative sentences, a bare adjectival predicate must be marked by *lao*:

- (59) ga ge dagaohaoqie \*(lao).  
 this CL cake tasty LAO  
 ‘This cake is tasty.’

In an alternative question with disjoined adjectival predicates, *lao* cannot appear in either disjunct:

- (60) ga ge dagao[haoqie (\*lao)] [fe haoqie (\*lao)]?  
 this CL cake tasty LAO NEG tasty LAO  
 ‘Is this cake tasty or not?’

If a *fe*-question is derived by the deletion of the second disjunct of (60), then it should also reject *lao*. This prediction, however, is not borne out:

- (61) ga ge dagao haoqie lao fe?  
 this CL cake tasty LAO FE  
 ‘Is this cake tasty or not?’



Thus, we do not assume that ellipsis derives *fe/me*-questions, but rather, *fe* and *me* are the negative polarities in DisjP; they appear sentence-finally because as a part of the DisjP, they are pied-pied by the null ‘whether’.

### 2.3. *Ma* and *a* end up in SpeechAct<sup>0</sup>

Section 2.1 has discussed two important differences between *fe/me*-questions and *ma*-questions. The first was that *fe/me*-questions but not *ma*-questions involve a polar DisjP, for which we have provided a syntactic analysis in section 2.2. The second difference was that *a*- and *ma*-questions cannot be embedded, but *fe/me*-questions without *a* can. This section provides an account of this fact by arguing that *ma* and *a* contribute the speech act force of requiring the addressee to answer the question (Krifka 2014; Holmberg 2016; Dayal 2025) and are in the SpeechAct (SA) head above CP.

We adopt the established view that a matrix question involves a higher SpeechAct Phrase (SAP) above CP in the left periphery (e.g., Rizzi 1997; Speas and Tenny 2003; Krifka 2014; Dayal 2025). A full root question that needs an answer from the addressee involves two semantic components (Hooper and Thompson 1973; Krifka 2014; Bhatt and Dayal 2020; Dayal 2025): a set of propositions encoded in the CP (Hamblin 1973), and a higher illocutionary operator, ASK (Dayal 2025) or QUEST (Krifka 2014), which we take to be SA<sup>0</sup> in this paper. This illocutionary operator puts the addressee under the obligation to assert which proposition in the set of propositions is true.

A key property of the illocutionary operator in questions is that it typically cannot be embedded by predicates like ‘know’ and ‘find out’, but can be embedded by predicates

like ‘wonder’ and ‘ask’.<sup>6,7</sup> Mandarin *ma*-questions have this property, leading us to posit that *ma*-questions have SA<sup>0</sup>. As we saw in (30), repeated below, *ma*-questions cannot be embedded by *zhidao* ‘know’, but it can be embedded by the predicate *wen* ‘ask’.

(30) \*wo zhidao Lisi xihuan he kafei ma.

I know Lisi like drink coffee MA

Intended: ‘I know whether Lisi likes to drink coffee.’

(62) Zhangsan wen wo Lisi xihuan kafei ma.

Zhangsan ask me Lisi like coffee MA

‘Zhangsan asked me whether Lisi whether likes coffee.’

In parallel, Yixing *a*-questions can be embedded under ‘ask’ (63), but not under ‘know’

(28), repeated below:

(28) wo xiaoze lao Zangsa huexi qie kafi {fe/\*fe a/\*fa}.

I know LAO Zhangsan like drink coffee FE/FE A/FA

‘I know whether Zhangsan likes to drink coffee.’

---

<sup>6</sup> The reason for this distribution is beyond the scope of our paper, but see Krifka (2014) and Dayal (2025) for the further explanation.

<sup>7</sup> While Krifka (2014) showed that speech act questions can be embedded by predicates like *ask* and *wonder*, Dayal (2025) argued that SAP only occurs in the matrix clause, and the embedded root question has an additional level, the Perspective Phrase that is above CP but below SAP. What they agreed on is that root questions involve an XP higher than CP, making them difficult to get embedded in most cases. We follow Krifka (2014) and Holmberg (2016) in taking this higher XP as SAP. This analysis is compatible with Dayal (2025) if this XP is Perspective Phrase.

(63) Zangsa      men   wo   Lisi   huexi   kafi      {fe a/fa}.

Zhangsan   ask   me   Lisi   like      coffee      FE A/FA

‘Zhangsan asked me whether Lisi likes coffee.’

We thus take these facts to indicate that *ma*- and *a*-questions involve SAP above CP.

Apart from the embedding pattern, an additional piece of evidence that *ma*- and *a*-questions involve SAP is their incompatibility with the SFP *ne* and its Yixing counterpart *nang*. Without *a*, *nang* can occur in *fe/me*-questions.

(64) a. *Mandarin ma-question*

Zhangsan   xihuan   he   kafei      ma      (\*ne)?

Zhangsan   like      drink coffee      MA      NE?

‘Does Zhangsan like to drink coffee?’

b. *Yixing questions*

Zangsa      huexi qie      kafi      {fe/\*fe a/\*fa/\*a}      nang?

Zhangsan   like   drink coffee      FE/FE A/FA/A      NANG

‘Does Zhangsan like to drink coffee?’

The SFPs *ne* and *nang* can appear sentence-finally in different types of matrix questions, including both yes/no questions and *wh*-constituent questions. While the syntactic details of these SFPs are not the focus of our paper, a common assumption is that Mandarin *ne* is located above C<sup>0</sup> (e.g., Paul 2015; Pan and Paul 2016; Soh 2025), presumably in SA<sup>0</sup>.

Assuming that there can only be one SA<sup>0</sup> in a clause, the incompatibility of *ma* and *a* with these SFPs suggests that *ma* and *a* are also in SA<sup>0</sup>, and thus they cannot all co-occur.

Since we argued in section 2.2 that *ma* is the interrogative C<sup>0</sup> in POLQs, we propose that *ma* moves from POLQ C<sup>0</sup> to the higher SA<sup>0</sup> via head movement. In other words, the functional item *ma* is the phonological exponent of both the POLQ interrogative feature and the SA feature and surfaces in SA<sup>0</sup>. Here we follow the Distributed Morphology framework (DM; e.g., Halle and Marantz 1993; Marantz 2006; Embick 2010; 2015) and traditional assumptions about head movement, where functional heads can be bundled together due to head movement and realised as a single functional item.

Since *ma* always marks yes/no questions, it has long since been treated as C<sup>0</sup> (c.f., Lee 1986; Paul 2015; Pan and Paul 2016, following Rizzi's 1997 split CP hypothesis). While we agree that *ma* is base-generated in C<sup>0</sup>, our approach differs from the previous studies in the following two ways. First, we argue that *ma* is C<sup>0</sup> exclusively associated with POLQs, to account for the lack of DisjP in *ma*-questions and the impossibility of *ma* in alternative questions and *wh*-constituent questions, a fact that the previous approaches that considered *ma* to be generic interrogative C<sup>0</sup> fail to account for. Second, we argue that *ma* moves from C<sup>0</sup> to SA<sup>0</sup>, to explain why *ma* can only appear in root questions, and why it cannot occur with the particle *ne* that is in a head position higher than C<sup>0</sup>.

We now turn to our analysis of the SFP *a* in Yixing. The fact that neither Mandarin *ma* nor Yixing *a* can be embedded or co-occur with other SA particles seems to suggest that Yixing *a* is also in SA<sup>0</sup>. We argue that *a* is base-generated in SA<sup>0</sup>, unlike *ma*, which moves to SA<sup>0</sup> from C<sup>0</sup>. Our argument is based on a key difference between *ma* and *a*: unlike *ma*, which cannot occur in *wh*-constituent questions or alternative questions, *a* can:

(65) a. ni huexi qie die a?

you like eat what A

‘What do you like eating?’

b. ni huexi qie kafi asi cuo a?

you like eat coffee or tea A?

‘Do you like to drink coffee or tea?’

This is because *a* is in SA<sup>0</sup> and selects an interrogative CP, regardless of whether that CP is an ALTQVN or a POLQ. In other words, *a* does not care whether the head of its complement is ALTQVN C<sup>0</sup> or POLQ C<sup>0</sup>. Thus, as predicted, *a* is not only compatible with ALTQVNs and *wh*-constituent questions, but also POLQs (i.e., yes/no questions without *fe* and *me*). We can use the diagnostics in section 2.1 to show that these yes/no questions without *fe* and *me* but with *a* are indeed POLQs. First, they can be answered by ‘It is so’:

(66) A: Zangsa huexi qie kafi de a?

Zhangsan like drink coffee DE A

‘Does Zhangsan like to drink coffee?’

B: si ge.

COP GE

‘It is so.’

Second, these questions can take a negative proposition:

(67) Zangsa fe huexi qie kafi a?

Zhangsan NEG like drink coffee A

‘Does Zhangsan not like to drink coffee?’

Our other diagnostics, the unconditional test and the ‘doubt’-test cannot apply to these questions because *a* cannot be embedded in general, as we saw in (28).

#### 2.4. Mandarin *bu/mei*-questions

Yixing *fe/me*-questions may remind Mandarin speakers of what we call *bu/mei*-questions in Mandarin (e.g., Zhang 1990; Cheng et al. 1996; C.-T. J. Huang et al. 2009), which insert negation at the end of a declarative. These questions were also called VP-NEG questions by C.-T. J. Huang et al. (2009). This subsection will show that *bu/mei*-questions behave like Yixing *fe/me*-questions in some aspects but like Mandarin *ma*-questions in others. This leads to our proposal for *bu/mei*-questions that is essentially a hybrid of our analyses of *fe/me*- and *ma*-questions.

We begin by discussing two similarities between Yixing *fe/me*-questions and *bu/mei*-questions. Like *fe/me*-questions, Mandarin *bu/mei*-questions end in an SFP that appears to be identical with the negations, *bu* and *mei* in the language. Like their counterparts in Yixing, negation *mei* and the corresponding SFP *mei* are used with perfective aspect, whereas negation *bu* and its corresponding SFP are used elsewhere.

(68) a. Zhangsan xihuan he kafei bu?

Zhangsan like drink coffee BU

‘Does Zhangsan like to drink coffee?’

b. Zhangsan jintian he-le kafei mei?

Zhangsan today drink-PERF coffee MEI

‘Did Zhangsan drink coffee today?’

Besides the insertion of negation at the end of a declarative, Mandarin *bu/mei*-questions share another similarity with Yixing *fe/me*-questions: they involve a polar DisjP, as can be shown with our diagnostics. As is observed by C.-T. J. Huang et al. (2009), *bu/mei*-questions cannot accept short answers like *shi de* ('It is so'):

(69) A: Zhangsan he-le kafei mei?

Zhangsan drink-PERF coffee MEI

Did Zhangsan drink coffee?'

B: #Shi de.

COP DE

'It is so.'

*Bu/mei*-questions cannot take a negative proposition, again suggesting that they involve a polar DisjP:

(70) \*Zhangsan bu xihuan he kafei bu?

Zhangsan NEG like drink coffee BU

Intended: 'Does Zhangsan not like to drink coffee?'

Given these similarities between *bu/mei*-questions and *fe/me*-questions, it may be tempting to assume that *bu/mei*-questions should have the same syntactic analysis as Yixing *fe/me*-questions. However, *bu/mei*-questions behave differently from *fe/me*-questions and similarly to Mandarin *ma*-questions in other aspects. The first is that unlike Yixing *fe/me*-questions but like Mandarin *ma*-questions, *bu/mei*-questions cannot be embedded (C.-T. J. Huang et al. 2009):

(71) \*wo bu xiaode [ta mai shu bu].

I NEG know he buy book BU

Intended reading: ‘I don’t know whether he buys books.’

(C.-T. J. Huang et al. 2009, 258)

Second, recall from section 2.3 that Mandarin *ma*-questions and Yixing *a*-questions cannot be followed by the SA SFP *ne* or *nang* (64). Neither can *bu/mei*-questions:

(72) \*Zhangsan xihuan he kafei bu ne?

Zhangsan like drink coffee BU NE

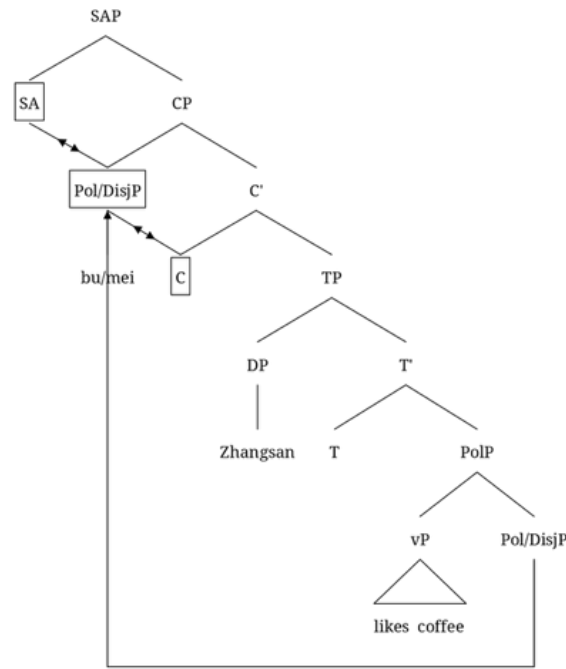
‘Does Zhangsan like to drink coffee?’

Our account of the incompatibility of *ma* and *a* with these SA SFPs was that *ma* and *a* are also in SA<sup>0</sup>, and there can only be one SA<sup>0</sup> in a sentence. Extending this account to Mandarin *bu/mei*, *bu* and *mei* are also in SA<sup>0</sup>.

Thus, on the one hand, *bu/mei*-questions pattern like Yixing *fe/me*-questions but not *ma*-questions in involving a polar DisjP. On the other hand, *bu/mei*-questions pattern like *ma*-questions but not *fe/me*-questions in being unembeddable and incompatible with the SFP *ne*. We take the former type of evidence as indication that *bu* and *mei* are the negative polarity in the polar DisjP, just like *fe* and *me*. We take the latter type of evidence as indication that *bu* and *mei* have SA features. Assuming that only heads can undergo head movement, we thus think that in *bu/mei*-questions, the polar DisjP in Spec, CP and C<sup>0</sup> fuse into a single head, and this head then fuses with SA<sup>0</sup> (73), resulting in the form of a single head.



(73)



Our analysis of *bu/mei*-questions involves base-generation of *bu/mei* in the TP domain and its movement to a higher position. This is also the assumption of Cheng et al. (1996) and Huang et al. (2009). There are two important differences between our analysis and this literature, however. First, while Cheng et al. did assume that negation is base generated in the TP domain, they did not explain why. For them, the negation is not part of a polar DisjP, but marks the negative polarity of a declarative sentence, but it is not clear why the yes/no question should be derived from a negative sentence, or how the negative sentence leads to the yes/no question meaning. In our analysis, the negation is base-generated in the TP because it is part of the DisjP. The other difference is that Cheng et al. took  $C^0$  to be the final landing site of negation, whereas we think its final

landing site is the higher SA<sup>0</sup>. Since C<sup>0</sup> should be allowed in embedded clauses, Cheng et al. could not explain why *bu/mei*-questions are unembeddable.

In sum, this section has argued that Yixing *fe/me*-questions involve a polar DisjP, whereas Mandarin *ma*-questions don't involve any DisjP or *wh*-movement. We then argued that *ma* moves from C<sup>0</sup> to SA<sup>0</sup>. Combining the properties of both *fe/me*- and *ma*-questions, *bu/mei*-questions involve both polar DisjP and fusion of C<sup>0</sup> and SA<sup>0</sup>.

Our analysis provides morphological evidence for Hamblin's (1973) question semantics, which claims there are two different interrogative C<sup>0</sup>s. POLQ C<sup>0</sup> is *ma* in Mandarin, whereas ALTQVN C<sup>0</sup> is null in Yixing. The fact that C<sup>0</sup> is null in A-not-A questions (to be discussed in section 3) and *wh*-constituent questions in Mandarin suggests that ALTQVN C<sup>0</sup> is also null in Mandarin. The two interrogative C<sup>0</sup>s also have different syntactic properties: only ALTQVN C<sup>0</sup> requires *wh*-movement; POLQ C<sup>0</sup> doesn't.

### **3. Whether the DisjP undergoes overt or covert *wh*-movement**

Having focused on Mandarin *ma*-questions and Yixing *fe/me*-questions so far, this section examines two other yes/no question constructions across the Chinese dialects—A-not-A questions and *a/ke*-questions. We will first apply our diagnostics of polar DisjP to these two question types, and show that they are ALTQVNs and involve a DisjP. But these two question types have a key difference from Yixing *fe/me*-questions and Mandarin *bu/mei*-questions, the other ALTQVNs that we have examined, in that the polar DisjP in A-not-A questions and *a/ke*-questions appears clause-medially rather than clause-peripherally. We show that despite appearing sentence-medially, this polar DisjP is sensitive to islands, leading us to posit the second parametric variation between *fe/me*-

questions on the one hand and A-not-A and *a/ke*-questions on the other: ‘whether’-movement is overt in the former but covert in the latter. This is important because we show that both overt and covert *wh*-movements are attested as the default question strategy in the Chinese language family, contrary to previous assumption. The (c)overtiness of ‘whether’-movement is a parameter across the Chinese dialects, in the same way that the (c)overtiness of *wh*-constituent-movement is between English and Mandarin Chinese.

We begin by introducing the background on A-not-A and *a/ke*-questions. Mandarin A-not-A questions are created by reduplicating the first syllable of the verb and inserting negation immediately after the reduplicant. As in *bu/mei*-questions, the negation inserted in A-not-A questions is sensitive to the aspect of the sentence, where *mei* is used with the perfective aspect, and *bu* is the negation elsewhere:

(74) a. Zhangsan xi bu xihuan he kafei?

Zhangsan RED NEG like drink coffee

‘Does Zhangsan like to drink coffee?’

b. Zhangsan he mei he kafei?

Zhangsan RED NEG drink coffee

‘Did Zhangsan drink coffee?’

As discussed with *bu/mei*-questions in section 2.4, negation in declaratives in Mandarin is sensitive to aspect. This leads us to think that *bu* and *mei* in A-not-A questions are genuine negations. Because multisyllabic words in Mandarin can sometimes be further divided into meaningful monosyllabic morphemes, some may not consider A-not-A

questions to involve phonological reduplication, but rather copying of a morpheme that happens to be monosyllabic. Following C.-T. J. Huang (1991), C.-T. J. Huang et al. (2009) and others, we believe A-not-A questions do involve genuine reduplication because it can reduplicate the first syllable of disyllabic words that cannot be further subdivided into morphemes (75a), the first syllable of loanwords like *PUA* that crucially cannot be further subdivided (*PUA* was adapted from English *pick-up artist*, and now means ‘abuse mentally, manipulate’ in colloquial Mandarin) (75b), and even in code-switching contexts (75c), where speakers’ reduplication patterns have been shown to respect sonority effects (Liu 2019).

(75) a. Zhangsan ma bu mahu?

Zhangsan RED NEG careless

‘Is Zhangsan careless?’

b. Zhangsan pi bu piyu’ei ta?

Zhangsan RED NEG PUA her

‘Does Zhangsan manipulate her?’

c. Zhe ge cai fre bu fresh?

this CL vegetable RED NEG fresh

‘Is this vegetable fresh?’

In addition to the first syllable of the verb, the entire verb or the VP can also be reduplicated (C.-T. J. Huang 1991). This section focuses on cases where only the first

syllable is reduplicated, and we assume that the other cases may also involve an ALTQVN C<sup>0</sup> and possibly a DisjP (e.g., V-, VP-, or TP-disjunction).<sup>8</sup>

There is another type of yes/no questions, what we call *a/ke*-questions, that have been attested in a variety of Chinese dialects, including varieties of Wu in Suzhou (Chao 1928), Taicang and Changshu, etc., Southwestern Mandarin in Kunming, Lower Yangtze Mandarin (also called Jianghuai Mandarin) in Nanjing and Hefei (e.g., Zhu 1985; Zhang 1990; C.-T. J. Huang 1991). In Suzhou, Taicang, Changshu and Nanjing, the morpheme *a* is inserted before the verb of a declarative sentence to create a yes/no question (7), repeated below. In Kunming and Hefei, the morpheme *ke* is inserted before the verb (76).<sup>9</sup>

(7) Zhangsan a xihuan huo kafei? *Nanjing*

Zhangsan A like drink coffee

‘Does Zhangsan like to drink coffee?’

(76) Zhangsan ke suong he kafei? *Hefei*

Zhangsan KE like drink coffee

‘Does Zhangsan like to drink coffee?’

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<sup>8</sup> It is possible that syntactic and postsyntactic processes can apply in this DisjP, such as ellipsis (C.-T. J. Huang 1991; C.-T. J. Huang et al. 2009) and Right Node Raising (Guo 2025).

<sup>9</sup> We mark the particles as *ke* and *a*, but their actual phonetic realisation varies slightly across dialects. For current purposes, we do not go into further phonetic detail here.

Recall from section 2 that we used four diagnostics of DisjP to contrast Mandarin *ma*-questions with Yixing *fe/me*-questions. We will now apply those diagnostics to A-not-A and *a/ke*-questions, and show that they involve a polar DisjP and are ALTQVNs.

First, recall that POLQs but not ALTQVNs can be answered by ‘so’ because ‘so’ requires the question to highlight only one proposition. Zhu (1985) observed that neither A-not-A questions like (74) nor *a/ke*-questions like (7) and (76) can be answered by *shi de* ‘It is so’ or their equivalents in the dialects. Below are more data from Nanjing:

(77) A: li     a   xihuan   huo     kafei (a)?

you   A   like     drink   coffee   A

‘Do you like to drink coffee?’

B: #Shi   dei.

COP   DEI

‘It is so.’

The second diagnostic we used to differentiate ALTQVNs and POLQs was embeddability by ‘doubt’, which requires its complement to highlight a single proposition. A-not-A and *a/ke*-questions can be embedded under verbs like ‘know’ (78), but not under ‘doubt’ (79).<sup>10</sup>

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<sup>10</sup> Y. Huang (2020) provided the opposite judgment, and claimed that ‘doubt’ can embed A-not-A questions:

(i) xiaohuanghuaiyi     xiaotang   shi     bu     shi     xihuan xiaoqiang.

Huang   suspect   Tang     RED   NEG   COP   like   Qiang

‘Huang suspects whether Tang likes Qiang.’

(Based on Y. Huang 2020:6)

(78) a. Lisi bu zhidao Zhangsan xi bu xihuan he kafei.

Lisi NEG know Zhangsan RED NEG like drink coffee

‘Lisi doesn’t know whether Zhangsan likes to drink coffee.’

b. Lisi bu xiaodei Zhangsan a xihuan huo kafei.

Lisi NEG know Zhangsan A like drink coffee

‘Lisi doesn’t know whether Zhangsan likes to drink coffee.’

c. Lisi bu xiaode Zhangsan ke suong he kafei.

Lisi NEG know Zhangsan KE like drink coffee

‘Lisi doesn’t know whether Zhangsan likes to drink coffee.’

(79) a. #Lisi huaiyi Zhangsan xi bu xihuan he kafei.

Lisi suspect Zhangsan RED NEG like drink coffee

Intended: ‘Lisi suspects whether Zhangsan likes to drink coffee.’

b. #Lisi huaiyi Zhangsan a xihuan huo kafei.

---

We think there is a confound here: *huaiyi* is ambiguous between ‘doubt’ and ‘be uncertain about’. The latter interpretation does not require its complement to highlight only one proposition, and can thus embed A-not-A questions. *Huaiyi* in (i) should be interpreted as ‘be uncertain about.’ To tease these two readings apart, we can build a context where ‘be uncertain about’ is not possible, such as the following, and here A-not-A is not possible:

(ii) Context: The police have found strong evidence that Zhangsan stole the book, and thus has reasonable suspicion that he did, though they cannot be fully certain.

#jingcha huaiyi Zhangsan tou mei tou shu.

police doubt Zhangsan RED NEG steal book

Intended: ‘The police suspects that Zhangsan stole the book.’

Lisi suspect Zhangsan A like drink coffee

Intended: 'Lisi suspects whether Zhangsan likes to drink coffee.'

c. #Lisi huaiyi Zhangsan ke suonghe kafei.

Lisi suspect Zhangsan KE like drink coffee

Intended: 'Lisi suspects whether Zhangsan likes to drink coffee.'

Third, given that ALTQVNs can be used as unconditionals, but POLQs cannot, the possibility of A-not-A questions and *a/ke*-questions in unconditionals suggests that they are ALTQVNs:

(80) a. buguan ni mingtian lai bu lai wo jia chi fan,

no.matter you tomorrow RED NEG come my home have meal

wo jintian dou yao mai henduo cai de.

I today DOU will buy many groceries DE

'Whether you're coming for a meal, I'll buy a lot of groceries today anyway.'

b. buguan li mingtian a lai wo jia chi fan,

no.matter you tomorrow A come my home have meal

wo jintian dou yao mai haoduo cai.

I today DOU will buy many groceries

'Whether you're coming for a meal, I'll buy a lot of groceries today anyway.'



c. buguan    ni    mingtian    ke    lai    wo    jia    chi    fan,  
 no.matter   you   tomorrow   KE   come my   home have meal  
 wo   jintian   dou   hui   mai   haoduo   cai.  
 I    today    DOU   will   buy   many   groceries

‘Whether you’re coming for a meal, I’ll buy a lot of groceries today anyway.’

Finally, recall that ALTQVNS cannot ask about a negative proposition because in these questions, Pol<sup>0</sup> has already been occupied by the polar DisjP. A-not-A and *a/ke*-questions require a positive proposition, again suggesting that they are ALTQVNS:

(81) a. \*Lisi xi    bu    bu    xihuan    he    kafei?

Lisi RED   NEG   NEG   like    drink coffee

Intended: ‘Does Zhangsan not like to drink coffee?’

b. \*Lisi a    bu    xihuan    he    kafei?

Lisi A   NEG   like    drink coffee

Intended: ‘Does Zhangsan not like to drink coffee?’

c. \*Lisi ke bu    suong    he    kafei?

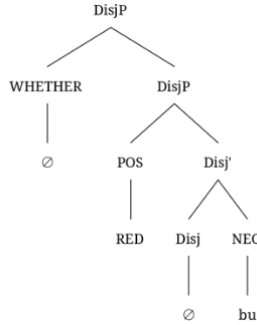
Lisi KE NEG   like    drink coffee

Intended: ‘Does Zhangsan not like to drink coffee?’

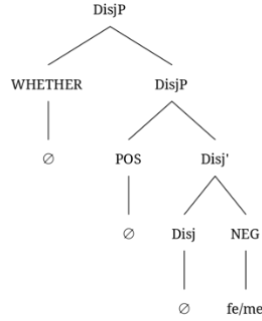
Therefore, we take these facts as evidence that A-not-A and *a/ke*-questions involve a polar DisjP. We think that the *A-not* part of A-not-A, *a* and *ke* are in fact the polar DisjP themselves. Section 4 will expand our view on *a* and *ke*, and here we present our view on the internal syntax of A-not-A. We think that *A-not* should be analysed as disjunction by a silent ‘or’ of the positive polarity, a reduplicant, and the negative polarity; a silent

‘whether’ is in Spec, DisjP (82). This parallels the structure of Yixing *fe/me* (83), whose positive polarity is also silent, and English *whether or not* (84), where only the positive polarity is silent.

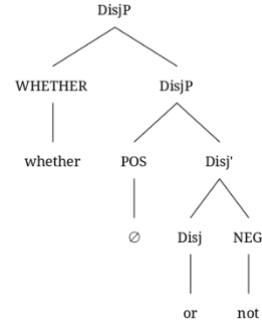
(82) *Structure of A-not*



(83) *Structure of fe/me*



(84) *Structure of or not*



We argue that in Mandarin A-not-A, the positive polarity reduplicates the verb to support the negative polarity, a prosodically weak particle.<sup>11</sup> This parallels *do*-support in English, but rather than inserting a dummy word *do*, Mandarin reduplicates the first syllable of the verb. Evidence for our analysis comes from other constructions across the Chinese dialects: in general, the Chinese dialects use reduplication as a strategy to support a prosodically weak particle. We see this strategy deployed not only in yes/no questions, but also in other constructions, leading to the A-Particle-A order. For example, in contrastive sentences in Shanghainese and Shanghai Mandarin, the first syllable of the

<sup>11</sup> Another possible analysis is verb doubling, where the verb moves, and multiple copies of the verb need to be pronounced, e.g. to support a bound morpheme (e.g. van Riemsdijk 1989; Abels 2000; Landau 2006) or due to morphological fusion (Nunes 2004; Cheng 2007; Corver and Nunes 2007). We do not pursue this analysis because there is no evidence that the verb has moved to the positive polarity (i.e. the first disjunct of the polar DisjP).

predicate under contrast is reduplicated, after which the prosodically light and toneless particle *me* is inserted (85).

(85) a. Zhangsan xi me xihuan de, tiao me tiaoti de.

Zhangsan RED ME like DE RED ME picky DE

‘Zhangsan does like her, but is still picky with her.’

b. Zhangsan ma me mahu de, shiqing zuo me daoshi zuowan de.

Zhangsan RED ME careless DE thing RED ME still finish DE

‘Zhangsan is careless, but he does finish his work.’

Likewise, in concessive sentences in Mandarin, the first syllable of the predicate is reduplicated, after which the focus marker *shi* is inserted (86).<sup>12</sup> We therefore think that the reduplication in A-not-A questions is an instantiation of the general strategy that the Chinese languages use to support a prosodically weak morpheme.

(86) a. zhe kafei, hao shi haohe, dan gui ye shi zhen gui.

this coffee RED SHI delicious but RED also SHI really expensive

‘This coffee does taste good, but it is also very expensive.’

---

<sup>12</sup> These concessive sentences are not to be confused with what Cheng and Vicente (2013) called verb doubling clefts, where a verb is clefted and both its copies are pronounced. In verb doubling clefts, the subject may intervene between the copula *shi* and the low copy of the verb, but this is not possible in concessive sentences:

(i) \*Ma shi Zhangsan mahu.

RED SHI Zhangsan careless

Intended: ‘As for being careless, Zhangsan is indeed careless.’

b. Zhangsan ma shi mahu-le dian, danshi ta mali ya.

Zhangsan RED SHI careless-LE a.little but he quick YA

‘Zhangsan is careless, but he is quick.’

Our analysis of A-not-A contrasts with C.-T. J. Huang (1991), C.-T. J. Huang et al. (2009) and Holmberg (2016), who saw the A-not-A morpheme together as a head with unvalued [Polarity]-feature, and this head just happens to look like negation. Our tests involving answerability by ‘so’ and embeddability under unconditionals and ‘doubt’ have suggested that A-not-A questions involve two propositions, one positive and one negative, not just one proposition with unvalued [Polarity]-feature as proposed in those previous studies. According to our analysis, it is not accidental that there is a negation-looking morpheme in A-not-A and Yixing *fe/me*-questions, but rather, this is the negative polarity that is disjoined with the positive polarity in the polar DisjP. This accounts for their sensitivity to aspect, just like negation in declaratives.

Having shown that A-not-A and *a/ke*-questions are ALTQVNs just like Yixing *fe/me*-questions, we note an obvious difference between them: the A-not-A morpheme, *a* and *ke* appear sentence-medially, whereas *fe* and *me* appear sentence-finally, which, as we have argued in section 2, is due to overt *wh*-movement. Following C.-T. J. Huang (1991), C.-T. J. Huang et al. (2009) and Holmberg (2016), we think that the A-not-A morpheme, *a* and *ke* also *wh*-move, but covertly. When embedded, these morphemes can lead to matrix questions, suggesting that they move long-distance from the embedded clause to the matrix Spec, CP.

- (87) a. Zhangsan xuancheng ta xi bu xihuan he kafei?  
 Zhangsan claim he RED NEG like drink coffee  
 ‘Does Zhangsan claim he likes to drink coffee or does he claim he doesn’t?’
- b. Zhangsan xuancheng ta a xihuan huo kafei?  
 Zhangsan claim he A like drink coffee  
 ‘Does Zhangsan claim he likes to drink coffee or does he claim he doesn’t?’
- c. Zhangsan xuancheng ta ke suong he kafei?  
 Zhangsan claim he KE like drink coffee  
 ‘Does Zhangsan claim he likes to drink coffee or does he claim he doesn’t?’

But when these morphemes are embedded in an island, the sentence becomes ungrammatical, suggesting that their movement is island-bound:

- (88) a. \*ta renshi-le [relative clause xi bu xihuan he kafei de xuesheng]?  
 he meet-PERF RED NEG like drink coffee DE student  
 Intended reading: ‘Did he meet a student who likes coffee, or did he meet a student who doesn’t like coffee?’
- b. \*ta renshi-le [relative clause a xihuan huo kafei de xuesheng]?  
 he meet-PERF A like drink coffee DE student
- c. \*ta renshi-le [relative clause ke suong he kafei de xuesheng]?  
 he meet-PERF KE like drink coffee DE student

(89) a. \*[<sub>subject</sub> Zhangsan xi bu xihuan he kafei] bijiao hao (ne)?

Zhangsan RED NEG like drink coffee more good NE

Intended reading: ‘Is it better that Zhangsan likes to drink coffee, or better that he doesn’t like to drink coffee’

b. \*[<sub>subject</sub> Zhangsan a xihuan huo kafei] bijiao hao (ne)?

Zhangsan A like drink coffee more good NE

c. \*[<sub>subject</sub> Zhangsan ke suong he kafei] bijiao hao (ne)?

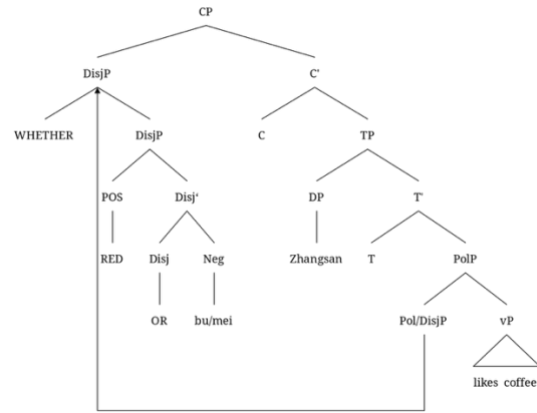
Zhangsan KE like drink coffee more good NE

These facts are based on C.-T. J. Huang’s (1991) and C.-T. J. Huang et al. ’s (2009) observations about A-not-A and Taiwanese *kam*-questions, the Taiwanese counterpart of *a/ke*-questions, which led them to propose that the A-not-A morpheme and *kam* are interrogative operators that move covertly to Spec, CP. We adopt their analysis of covert movement, but in our analysis, the A-not part is polar DisjP. What actually moves is the covert ‘whether’ that is adjacent to the polar DisjP; because its movement is covert, we do not know whether it covertly pied-pipes the polar DisjP.<sup>13</sup> Following is the tree where the covert ‘whether’ moves and pied-pipes the DisjP covertly:

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<sup>13</sup> Since this ‘whether’ is covert, an alternative analysis is that it moves overtly as ‘whether’ does in Yixing, except that ‘whether’ does not pied-pipe the polar DisjP in A-not-A and *a/ke*-questions. While this analysis is in principle possible given the examples we have seen so far, section 4 will discuss colloquial Singapore Teochew, where *ke*’s counterpart clearly has the *wh*-feature. This *wh*-element stays in-situ and sentence-medially, but is subject to island effects, suggesting that ‘whether’ moves covertly rather than overtly.

(90)



The same analysis of covert movement applies to *a/ke*-questions. Section 4 will show that *a* and *ke* realise the entire DisjP, including ‘whether’. Since they include the *wh*-feature, they undergo covert-movement to Spec, CP.

To summarise, after section 2 discussed the first parameter of variation in yes/no questions (i.e., whether a yes/no question involves a polar DisjP), this section has introduced the second parameter—in ALTQVNs, whether ‘whether’ moves overtly or covertly. We use two new yes/no question constructions, A-not-A questions and *a/ke*-questions, to show that while they involve a DisjP, they involve covert *wh*-movement which can move long-distance, but not across an island. This contrasts with Yixing *fe/me*-questions, which involve overt *wh*-movement. We agree with C.-T. J. Huang (1991), C.-T. J. Huang et al. (2009) and Holmberg (2016) that A-not-A and *a/ke*-questions involve covert *wh*-movement, but differ in the internal structure of A-not-A, and what is the *wh*-element.

#### 4. Whether the DisjP is realised as a single VI or multiple ones

This section discusses the third parameter of variation across the Chinese dialects—in ALTQVNS, whether the polar DisjP is realised as a single VI or multiple ones. As we have claimed in the previous section, the DisjP is spelt out as multiple VIs in A-not-A questions. They will be contrasted with *a* and *ke* in subsection 4.1, which realise the DisjP in a single VI. In subsection 4.2, we discuss the fact that the interaction of the second and third binary parameters predicts there to be a type of yes/no question with overt *wh*-movement and realisation of the DisjP as a single VI. Then we show that *bo*-questions in Liangshan dialect are precisely this.

##### 4.1. *A* and *ke* realise the DisjP as a single VI

The previous sections have shown that *fe/me*-, A-not-A and *a/ke*-questions all involve a polar DisjP. This DisjP is spelt out as multiple VIs, including some silent ones in *fe/me*- and A-not-A questions. We think *a* and *ke* are most likely to be the DisjP in *a/ke*-questions, and the question is whether they are part of a multi-VI sequence including silent VIs (as in *fe/me*-questions), or they are a single VI. We argue for the latter because unlike the negation in A-not-A and Yixing *fe/me*-questions, the morphology of *a* and *ke* is not sensitive to the aspect of the sentence. They remain *a* and *ke*, regardless of whether the sentence aspect is perfective (91b) or not (91a).

(91) a. li      a/ke xihuan he kafei?

you A/KE like drink coffee

‘Do you like drinking coffee?’



b. li      shangwu    a/ke   he-le      na-bei   kafei?

you   morning    A/KE   drink-PERF   that-cup   coffee

‘Did you drink that cup of coffee this morning?’

We thus propose that the DisjP is realised as a single VI in *a/ke*-questions, specifically, the features of the DisjP are packaged into a single functional morpheme (i.e., the particle *a* or *ke*) in Late Insertion (see (94a-b) later).<sup>14</sup>

Furthermore, we argue that the features expounded by *a* and *ke* also include the *wh*-feature of ‘whether’ in Spec, DisjP because *a* and *ke*’s counterpart in colloquial Singapore Teochew, *ka*, has been reported to be able to occur alone (92), or co-occur with A-not-A (93).<sup>15</sup>

(92) Ah Meng   ka    suka   ji    bun   zi?

Ah Meng   KA    like   this   CL    book

‘Does Ah Meng like this book?’

(Cole and Lee 1997:190)

(93) Ah Meng   ka    su    m    suka   ji    bun   zi?

Ah Meng   KA    RED   NEG   like   this   CL    book

‘Does Ah Meng like this book?’

(Cole and Lee 1997:192)

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<sup>14</sup> This is in line with Bobaljik’s (2012) analysis of portmanteaus, which are the result of Vocabulary Insertion ‘at a node dominating multiple features corresponding to multiple terminal nodes at some state of the derivation’ (Bobaljik 2012:148). Studies not specifically in the DM framework have also posited realising the features of an XP with a single VI (e.g., Neeleman and Szendrői 2007).

<sup>15</sup> Our Singapore Teochew speaker consultant found the co-occurrence of *ka* and A-not-A to be unnatural, suggesting there is inter-speaker variation in this.

We take this to mean that *ka* may either realise the entire polar DisjP, including the *wh*-element ‘whether’ as in (92), or only realise ‘whether’ as in (93).<sup>16</sup> In the latter case, the A-not-A form is the rest of the DisjP.<sup>17</sup> In both cases, *ka* realises the *wh*-feature. Thus, we assume that its counterparts *a* and *ke* also realise the *wh*-feature. As far as we know, *a* and *ke* cannot co-occur with A-not-A in all the dialects discussed in this paper (Zhu 1985), as was confirmed by our consultants, suggesting that *a* and *ke* must expone the entire polar DisjP rather than ‘whether’ alone.

It is worth mentioning that *ka* constitutes an indirect argument that *a* and *ke* move covertly. As Cole and Lee argued, *ka* undergoes covert *wh*-movement because when it is embedded in an island, we lose the relevant yes/no reading. If *ka*’s counterparts *a* and *ke* are single VIs that include the *wh*-feature, then their sentence-medial position and sensitivity to islands suggest that they also move covertly like *ka*.

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<sup>16</sup> This is a simplification of Cole and Lee’s (1997) analysis that *ka* is  $Q^0$ , and there’s a covert *wh*-element in Spec, *ka*P that moves covertly.

<sup>17</sup> We think that the so-called A-not-A form in colloquial Singapore Teochew is not equivalent to the A-not-A form in Mandarin discussed in section 3. According to Cole and Lee (1997), the A-not-A form in colloquial Singapore Teochew can appear alone without *ka*; and when it does, it can be embedded in islands and still give rise to the relevant yes/no reading, in contrast to *ka*. We observe that this parallels the island insensitivity of alternative questions with transparent disjunction by *haishi* ‘or<sub>whether</sub>’ in Mandarin (footnote 5), and thus think the A-not-A form in colloquial Singapore Teochew may be a transparent DisjP, in line with a similar point by Cole and Lee. In contrast, the island sensitivity of Mandarin A-not-A questions suggests that they involve covert *wh*-movement.

In contrast to *a/ke*-questions, in *fe/me*- and A-not-A questions, the polar DisjP is phonologically realised with distinct VIs. We elaborate the trees in (82) and (83) further in (94) to which we add *a/ke*-questions. As will be discussed in detail in section 5, this variation in (94) is consonant with some versions of the Borer-Chomsky Conjecture (BCC) that attribute crosslinguistic variation to different ways to realise the same features morphophonologically (e.g., Borer 2013).<sup>18</sup>

(94) *How polar DisjP is realised*

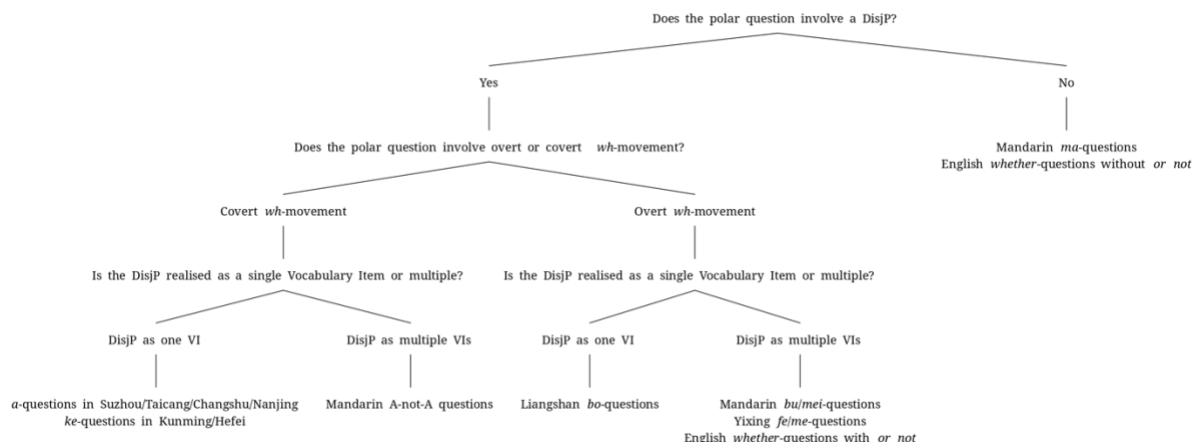
|           |                          |                         |
|-----------|--------------------------|-------------------------|
| <i>a.</i> | <i>a</i>                 | <i>Nanjing, etc.</i>    |
| <i>b.</i> | <i>ke</i>                | <i>Hefei, etc.</i>      |
| <i>c.</i> | <WHETHER> <POS> <OR> NEG | <i>Yixing</i>           |
| <i>d.</i> | <WHETHER> RED <OR> NEG   | <i>Mandarin A-not-A</i> |
| <i>e.</i> | whether <POS> or not     | <i>English</i>          |

Following is a more complete picture of parametric variation, repeated from (9), including all three parameters—(a) whether the yes/no question involves a DisjP; (b) whether the yes/no question involves overt or covert *wh*-movement; and (c) how the DisjP is realised morphologically.

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<sup>18</sup> DM-based studies do not explicitly discuss BCC or deep reasons behind variation, but we think the DM tradition is consistent with it. As Embick (2015:32) proposed, ‘languages may differ in terms of how they package their features into functional morphemes.’ Here, Embick’s feature packaging precisely concerns what feature(s) can be realised by a single functional item.

## (9) Parameters of Yes/no Questions



The second parameter, whether a yes/no question involves overt or covert *wh*-movement, is logically independent from the third parameter, how the DisjP is realised morphologically. Among covert movement constructions, we have seen variation along the third parameter: Mandarin *A-not-A* questions realise the DisjP as multiple VIs, whereas *a/ke*-questions realise the DisjP as a single one. We should expect this variation among overt movement constructions as well, where some realise the DisjP as multiple morphemes, as in Yixing *fe/me*-questions, while others realise it as a single VI that undergoes overt movement. We discuss precisely this possibility in the next subsection.

### 4.2. DisjP as a VI plus overt *wh*-movement: The case of Liangshan dialect

*Bo*-questions in Liangshan dialect of Southwestern Mandarin, spoken in Liangshan area of China's Sichuan Province, involve overt *wh*-movement and realisation of DisjP as a single VI. To the best of our knowledge, *bo*-questions have never been discussed in the literature before.

These questions end with the SFP *bo*, which is phonologically similar to the non-perfective negation marker *bu* in this dialect. Given this resemblance, it may be tempting to analyse this question form in parallel with Yixing *fe/mei*-questions or Mandarin *bu/mei*-questions, whose SFPs are negation. However, there is evidence that *bo* is not negation. As in Mandarin and Yixing, sentence negation in Liangshan is sensitive to the aspect: *bu* is used with the imperfective aspect (95a), and *mei* is used in perfective situations (95b).

(95) a. Zhangsan bu xihuan he kafei.

Zhangsan NEG like drink coffee

‘Zhangsan does not like to drink coffee.’

b. Zhangsan mei he kafei.

Zhangsan NEG drink coffee

‘Zhangsan didn’t drink coffee.’

In contrast, the yes/no question marker *bo* is invariant, regardless of aspect:

(96) a. Zhangsan xihuan he kafei bo?

Zhangsan like drink coffee BO

‘Does Zhangsan like to drink coffee?’

b. Zhangsan he-le kafei bo?

Zhangsan drink-PERF coffee BO?

‘Has Zhangsan drunk coffee?’

It may thus appear that *bo* is like *ma* in Mandarin—they both mark a yes/no question in the sentence-final position, and do not change their form based on aspect. However, when we

apply the four tests of polar DisjP, *bo*-questions in Liangshan pattern do involve a DisjP.

First, they reject the answer *shi de* ('It is so'):

(97) A: Zhangsan xihuan he kafei bo?

Zhangsan like drink coffee BO

'Does Zhangsan like to drink coffee?'

B: #Shi de.

COP DE.

'It is so.'

Second, *bo*-questions can generally be embedded under verbs like 'know', but cannot be embedded under 'doubt':

(98) a. wo bu xiaodei Zhangsan xihuan he kafei bo.

I NEG know Zhangsan like drink coffee BO

'I don't know whether Zhangsan likes to drink coffee.'

b. #Lisi huaiyi Zhangsan xihuan he kafei bo.

Lisi suspect Zhangsan like drink coffee BO

Intended: 'Lisi suspects whether Zhangsan likes to drink coffee.'

Third, *bo*-questions can occur in unconditionals:

(99) buguan ni mingtian lai wo jia chi fan bo,

no.matter you tomorrow come my home have meal BO

wo jintian dou hui mai haoduo cai.

I today DOU will buy many groceries

'Whether you're coming for a meal, I'll buy a lot of groceries today anyway.'

Finally, *bo*-questions do not allow negative propositions:

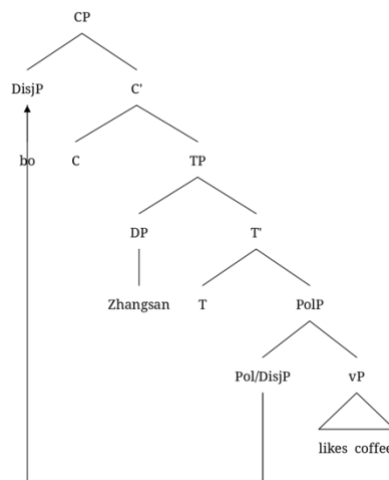
(100) \*Zhangsan bu xihuan he kafei bo?

Zhangsan NEG like drink coffee BO

Intended: ‘Does Zhangsan not like to drink coffee?’

The empirical data suggest that *bo*-questions behave like Mandarin *ma*-questions in some ways but like Yixing *fe/me*-questions in others. First, (95) suggests that Liangshan *bo*, like Mandarin *ma* but unlike Yixing *fe/me*, is not negation because it is not sensitive to aspect. Then (97)–(100) suggest that *bo*-questions, like *fe/me*-questions but unlike *ma*-questions, involve a DisjP. Their general embeddability suggests that *bo*-questions do not involve SA<sup>0</sup>. Taken together, these facts lead us to propose that the polar DisjP in *bo*-questions is realised as a single VI *bo*, just like *a* and *ke*. Since it realises the entire DisjP, not just the negative polarity in it, it is not sensitive to aspect. *Bo*’s sentence-final position, we argue, is due to the overt *wh*-movement of *wh*-DisjP (101).

(101)



Liangshan thus completes the paradigm: *bo*-questions externalise the DisjP in a single VI as in Nanjing and Hefei *a/ke*-yes/no questions, but involve overt *wh*-movement as in Yixing *fe/me*-questions.

## 5. Conclusion

This paper has surveyed an apparently diverse set of nine yes/no question constructions across English and the Chinese variants, and argued that they are in fact complete systematic variation generated by just three binary parameters, which we summarise with the parameter hierarchy in (9). This apparent diversity of yes/no questions is due to their unique intersection of alternatives and question formation—on the one hand, questions may be formed with or without syntactic generation of alternatives; on the other hand, questions may vary in how those alternatives are morphologically realised, and whether they involve overt or covert movement.

As we lay out these three parameters, we have made two main theoretical contributions. The first is for question semantics and typology. We have provided morphological support for Hamblin's (1973) proposal of two distinct interrogative C<sup>0</sup>s, one in ALTQVNS and the other in POLQS. Not only are they realised differently, but we have also shown that they are associated with different syntactic properties, where ALTQVN C<sup>0</sup> requires (overt or covert) *wh*-movement, but POLQ C<sup>0</sup> doesn't. The fact that POLQS do not involve any *wh*-element, not even genuine *wh*-in-situ, enriches the typology of questions and suggests that there are at least three question types across languages.

Our second contribution is for the theories of parametric variation. Many different but interrelated approaches to parametric variation have been proposed in the Minimalist



Program (for a recent comprehensive study, see Roberts 2019). We divide them into three types—two different interpretations of ‘Borer-Chomsky Conjecture (BCC)’, and what we call the ‘Chomskyan Externalisation’. We describe each approach in more detail below, and argue that the intricate cross-linguistic and cross-dialectal variation uncovered in this paper supports all three approaches, with each parameter being accounted for by one approach.

Perhaps the best-known approach to parametric variation in minimalism is BCC (e.g., Baker 2008; Roberts 2019), which attributed variation to the lexicon. Less explicitly stated in the literature is the fact that there are actually two different versions of BCC. The first is what we call the ‘Feature Availability’ version, where cross-linguistic variation is the result of the presence or absence of certain features in the lexicon. Specifically, Chomsky (2001) argued that variation occurs because a language makes a one-time selection of the universal set of formal features before it builds up its lexicon, and different languages may choose different sets of features. This idea was supported using a wide range of cross-linguistic data in major parametric studies such as Biberauer et al. (2014) and Roberts (2019). Following the ‘Feature Availability’ version of the BCC, our first parameter (i.e., whether a language has POLQ or ALTQVN or both) is due to what features a language has chosen for its  $C^0$ .

The second version of BCC is Borer’s (2013), which focused on morphophonological realisation of features. She attributed cross-linguistic variation to what she called ‘realisational properties’, where the same functional item or feature might be realised by different surface phonological forms (Borer 2013:630). We call this the ‘Feature

Realisation’ version of the BCC, to be contrasted with the more widely adopted ‘Feature Availability’ version that we introduced above. Our third parameter (i.e., whether the DisjP is realised as a single VI or multiple ones) is precisely variation in features’ morphophonological realisation and thus supports the ‘Feature Realisation’ version of the BCC.

There is a series of recent works under the ‘Chomskyan Externalisation’ approach that argued that cross-linguistic variation should be restricted to externalisation (Berwick and Chomsky 2016; Chomsky 2019; Chomsky et al. 2023): there is no variation in the Merge-based narrow syntax computation, and variation happens when the universal hierarchical structure is externalised at the sensorimotor (SM) interface. In some way, the ‘Feature Realisation’ version of the BCC may be subsumed under this approach because morphophonological realisation is a part of externalisation. But the works in the ‘Chomskyan Externalisation’ approach have primarily focused on another aspect of externalisation, linearisation, such as head directionality and copy spell-out. Our second parameter (i.e., whether a yes/no question involves overt or covert *wh*-movement) is precisely variation in linearisation—which *wh*-copy in the *wh*-movement chain is pronounced. Furthermore, we would like to argue that the ‘Feature Realisation’ version of BCC and the Chomskyan Externalisation not only share the same core that variation is restricted to externalisation, but also complement each other: with the former approach focusing on the lexicon, and the latter on word order, together they provide a full picture of how externalisation may lead to parametric variation.

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